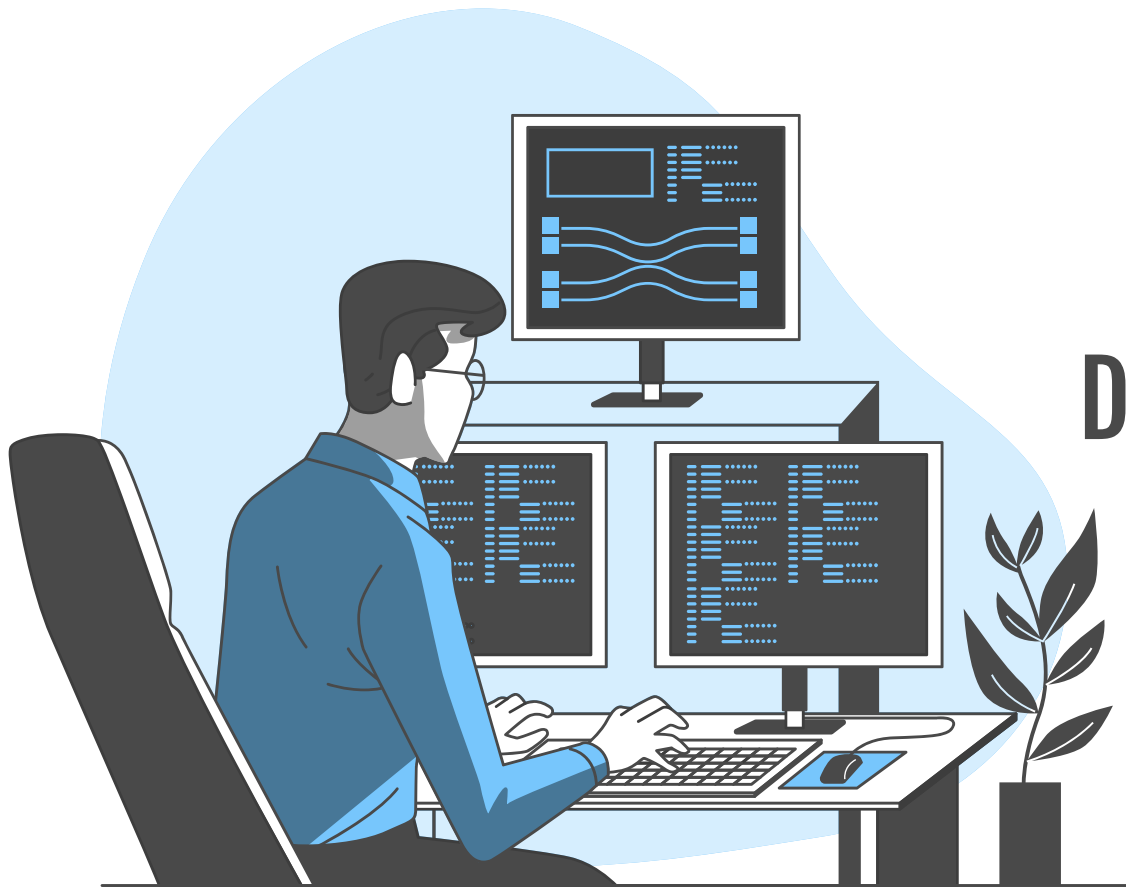
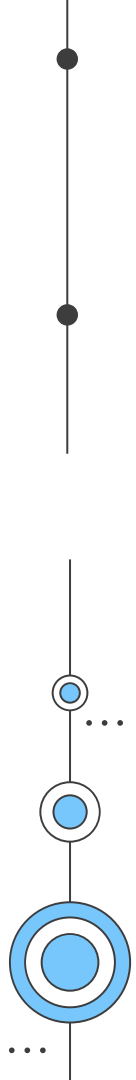




AI Assisted Development and Git Mastery

Dorin Daniel Radulescu
Toea



01

Cooperation With AI

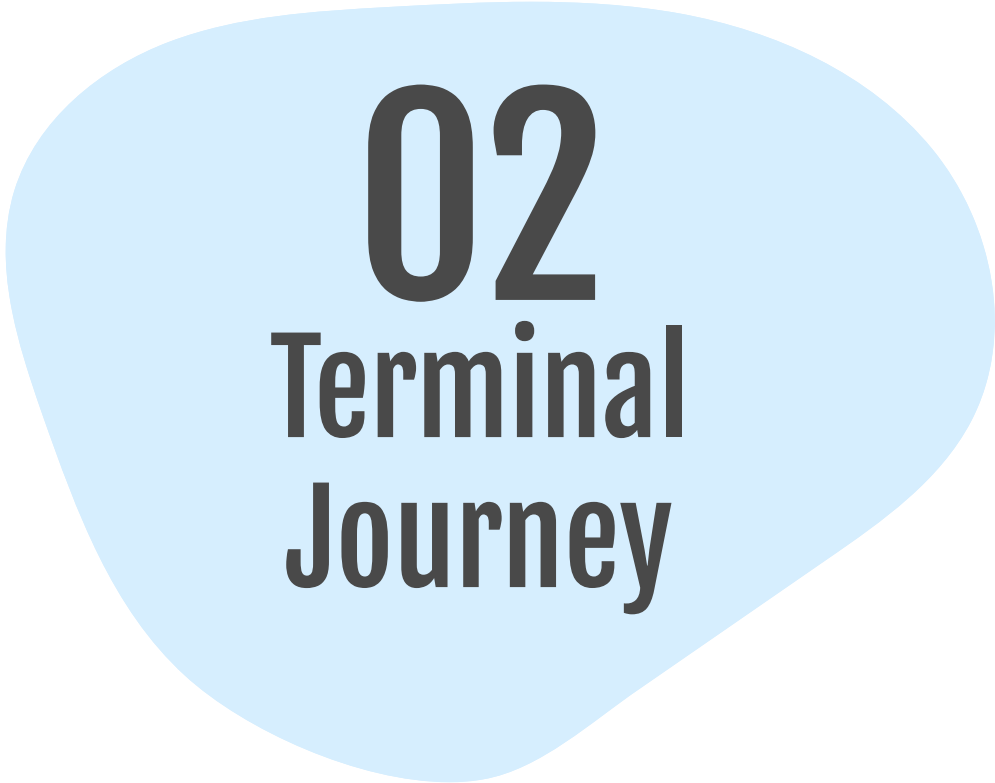
For this little project, i used Gemini AI to
build a program for expense management.
The prompt used was this:



hazme un pequeño programa de un gestor de gastos en java,
añadiendo una rama en la que se añada un fragmento de
código en el que se calcule el total gastado hasta el momento

...



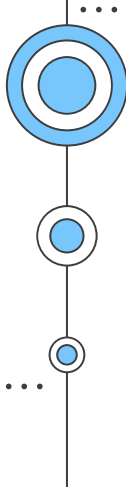


02

Terminal Journey



...



To initialize the project on Visual Studio Code, we should make the directory on the local system using `<mkdir 'project name'>`.



```
PS C:\Users\Usuario> mkdir dorindaniel-modulo20
```

Directorio: C:\Users\Usuario

Mode	LastWriteTime	Length	Name
d----	11/05/2026	9:57	dorindaniel-modulo20

```
PS C:\Users\Usuario> cd dorindaniel-modulo20
PS C:\Users\Usuario\dorindaniel-modulo20>
```



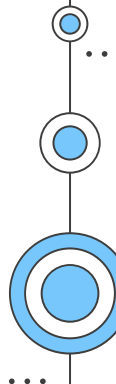
Then, we move to the directory by using `<cd 'project name'>`.

To initialize, we use `<git init>`



```
PS C:\Users\Usuario\dorindaniel-modulo20> git init
Initialized empty Git repository in C:/Users/Usuario/dorindaniel-modulo20/.git/
PS C:\Users\Usuario\dorindaniel-modulo20>
```

...



For adding an existing repository from GitHub, we must use `<git remote add origin 'url'>`.



```
PS C:\Users\Usuario\dorindaniel-modulo20> git remote add origin https://github.com/lordo0174/ai-git-mastery.git
PS C:\Users\Usuario\dorindaniel-modulo20> |
```

```
PS C:\Users\Usuario\Desktop\modulo20> git add .
```



Then, we add the file changes with `<git add .>`.

To let people know about the changes, we use `<git commit -m "message">`.



```
PS C:\Users\Usuario\Desktop\modulo20> git commit -m "V 1.0"
[main (root-commit) aae6091] V 1.0
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 GestorGastos.java
```

...

To push the changes on the GitHub repository, we use `<git push -u origin main>`.



```
PS C:\Users\Usuario\Desktop\modulo20> git push -u origin main
Enumerating objects: 3, done.
```

The code looks like this:



```
1 package com.usuarios;
2 import java.util.ArrayList;
3 import java.util.Scanner;
4
5 public class GestorGastos
6 {
7     //Constructor
8     public static void main(String[] args)
9     {
10         ArrayList<String> conceptos = new ArrayList<>();
11         ArrayList<Double> montos = new ArrayList<>();
12         Scanner sc = new Scanner(System.in);
13         boolean salir = false;
14
15         while (!salir)
16         {
17             System.out.println("\n--- MI GESTOR DE GASTOS ---");
18             System.out.println("1. Añadir gasto");
19             System.out.println("2. Ver lista de gastos");
20             System.out.println("3. Salir");
21             System.out.println("Elige una opción: ");
22
23             int opcion = sc.nextInt();
24             sc.nextLine(); // limpiar buffer
25
26             if (opcion == 1)
27             {
28                 System.out.print("Concepto: ");
29                 conceptos.add(sc.nextLine());
30                 System.out.print("Monto: ");
31                 montos.add(sc.nextDouble());
32                 System.out.println("Gasto guardado!");
33             }
34             else if (opcion == 2)
35             {
36                 System.out.println("\n--- lista de Gastos ---");
37                 for (int i = 0; i < conceptos.size(); i++)
38                 {
39                     System.out.println(i + " - " + conceptos.get(i) + " : $" + montos.get(i));
40                 }
41             }
42             else if (opcion == 3)
43             {
44                 salir = true;
45                 System.out.println("Cerrando programa...");
46             }
47             sc.close();
48         }
49     }
50 }
```

For creating a new branch named feature-update, we use <git checkout -b feature-update>.

```
System.out.println(x: "4. Calcular total gastado");  
else if (opcion == 4)  
{  
    double total = 0;  
    for (double monto : montos)  
    {  
        total += monto;  
    }  
    System.out.println(x: "\n--- Total Gastado ---");  
    System.out.println("El total gastado es: $" + total);  
}
```



```
PS C:\Users\Usuario\Desktop\modulo20> git checkout -b feature-update  
Switched to a new branch 'feature-update'
```



On the new branch, we insert the new code fragment.

...

...
...
...
We use `<git add .>` and `<git commit -m "message">` to upload the changes on the branch `feature-update`.



```
PS C:\Users\Usuario\Desktop\modulo20> git add .  
PS C:\Users\Usuario\Desktop\modulo20> git commit -m "V 1.1"  
[feature-update c37b59d] V 1.1  
1 file changed, 59 insertions(+)
```

```
PS C:\Users\Usuario\Desktop\modulo20> git checkout main  
Your branch is up to date with 'origin/main'.  
PS C:\Users\Usuario\Desktop\modulo20> git merge feature-update  
Updating aae6091..c37b59d  
Fast-forward  
GestorGastos.java | 59 ++++++  
1 file changed, 59 insertions(+)
```



Then we change to the main branch and do a merge using `<git checkout main>` and `<git merge feature-update>`.

To finish, we use `<git push -u origin main>` to push the final changes to the GitHub repository.



```
PS C:\Users\Usuario\Desktop\modulo20> git push -u origin main  
Enumerating objects: 5, done.  
Counting objects: 100% (5/5), done.  
Delta compression using up to 12 threads  
Compressing objects: 100% (2/2), done.  
Writing objects: 100% (3/3), 856 bytes | 856.00 KiB/s, done.  
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)  
To https://github.com/lordo0174/ai-git-mastery.git  
   aae6091..c37b59d  main -> main  
branch 'main' set up to track 'origin/main'.
```

...



03

Program Test



...



To show the program works properly, we run the Java code.



```
--- MI GESTOR DE GASTOS ---
1. Añadir gasto
2. Ver lista de gastos
3. Salir
4. Calcular total gastado
Elige una opción: 1
Concepto: Pago modelo 303 2T
Monto: 256
¡Gasto guardado!

--- MI GESTOR DE GASTOS ---
1. Añadir gasto
2. Ver lista de gastos
3. Salir
4. Calcular total gastado
Elige una opción: 4

--- Total Gastado ---
El total gastado es: $256.0

--- MI GESTOR DE GASTOS ---
1. Añadir gasto
2. Ver lista de gastos
3. Salir
4. Calcular total gastado
Elige una opción: 2

--- Lista de Gastos ---
- Pago modelo 303 2T: $256.0
```

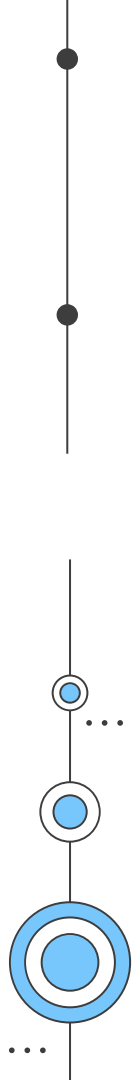
...





03

GitHub Pages



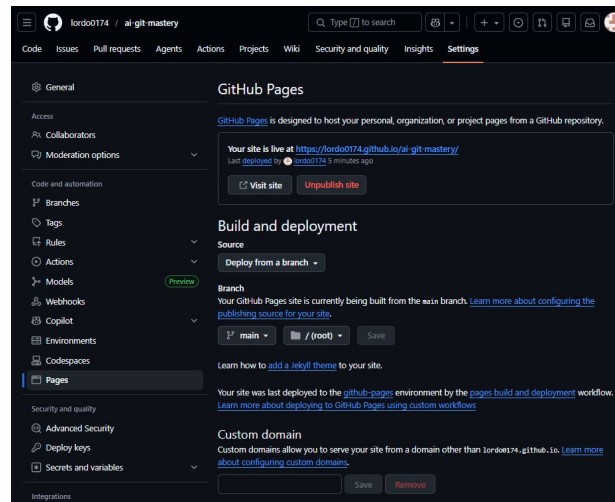
For the GitHub Pages, we need to do an index.html for display the documentation of the project on a website.

When we done finished the html code, we use `<git add .>`, `<git commit -m "message">` and `<git push origin main>`.

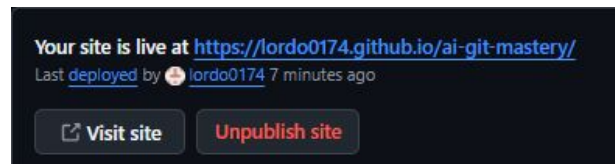
```
PS C:\Users\Usuario\Desktop\modulo20> git add index.html
PS C:\Users\Usuario\Desktop\modulo20> git commit -m "Added documentation for GitHub Pages"
[main 7037561] Added documentation for GitHub Pages
 1 file changed, 126 insertions(+)
 create mode 100644 index.html
PS C:\Users\Usuario\Desktop\modulo20> git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 1.75 KiB | 1.75 MiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/lordo0174/ai-git-mastery.git
 c37b59d..7037561  main -> main
```

...

To get this index.html on a public url, we need to go to settings on the GitHub Repository and open Pages section.



Then, we use the branch main, select the root and we done saving. After 1-2 minutes, we already have a public URL with out index.html.

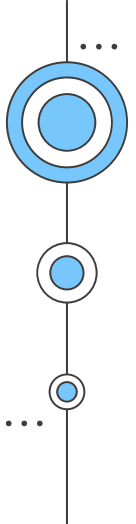




04

Deployment and repository

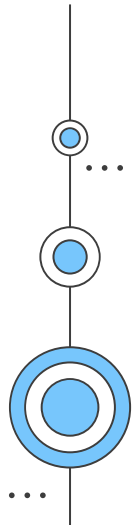


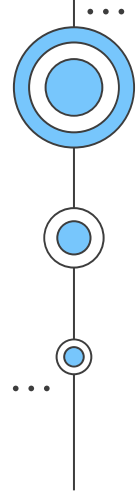


The public link of my repository is this:

<https://github.com/lordo0174/ai-git-mastery.git>

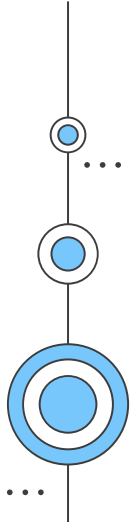
...





05

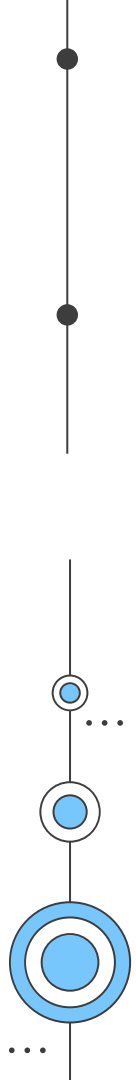
Conclusions and Cheat Sheet





Through this project, I have mastered the Feature Branch Workflow, which is the industry standard for collaborative software development. Learning to isolate new features in dedicated branches before merging them into the main line ensures code stability and project integrity. This assignment demonstrated that version control is not just about saving files, but about managing the entire lifecycle of a digital product.

The link of my GitHub Pages with the Cheat Sheet is:
<https://lordo0174.github.io/ai-git-mastery/>



...

